Exploring Trade-offs in Explainable AI

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Abstract

Machine Learning (ML) models are increasingly used in systems that involve physical human interaction or decision-making systems that impact human health and safety. Ensuring that these systems are safe and reliable is an important topic of current AI research.

For many ML models it is unclear how a prediction (output) is arrived at from the provided features (input). Critical systems cannot blindly trust the predictions of such "black box" models, but instead need additional reassurance via insight into the model's reasoning. A range of methods exist within the field of Explainable AI (XAI) to make the reasoning of black box ML models more understandable and transparent.

The explanations provided by XAI methods may be evaluated in a number of (competing) ways. In this paper, we investigate the trade-off between selected metrics for an XAI method called UnRAvEL, which is similar to the popular LIME approach. Our results show that by weighting the terms within the acquisition function used in UnRAvEL, different trade-offs can be achieved.

Keywords—Machine Learning, Explainable AI, Gaussian Process, LIME, UnRAvEL.

1 Introduction

There is increasing demand for machine learning (ML) models to be used within critical systems. One of the issues holding this back is the understanding of the decision making, and the proof of reliability of these models. Current high performing ML models are commonly complex black box models. With these models it is difficult to understand how the outputs of these are achieved, and the importance of each feature in determining the output.

Figure 1 shows a typical model, which takes as input a vector X of input features (x_1, x_2, …, x_n) and generates a prediction f(X) that depends on X. But as we have little or no insight into the reasoning of each prediction, the model is described as being "black box".

Explainable AI (XAI) is a discipline that is attempting to provide insight into the reasoning of black box models [1]. One common XAI approach is to identify the contribution of each feature in the determination of a model's prediction.

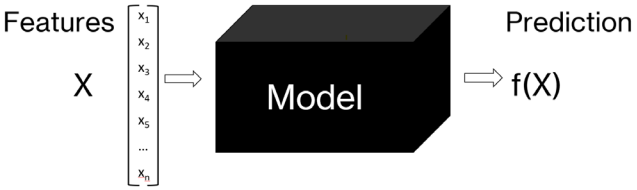

Figure 1. A typical black box model with opaque reasoning

One of main XAI approaches is LIME (Local Interpretable Model-agnostic Explanations) [2]. In the original LIME paper there is a simple experiment to demonstrate why XAI is needed.

Figure 2. An image from the LIME [2] paper. The left image is the image presented as test data for the model, it is of a husky with a background of snow. The image on the right shows the areas of the image identified by LIME that were the most important in the prediction

In the experiment, Figure 2, a model is built to distinguish between images of wolves and images of huskies. The model is built, and it achieves a good performance. However, it has been deliberately trained to be a flawed model. All the training images of wolves presented to the model at training have snow in the background, and all images of huskies do not have snow. The model is then presented with the test images, and all test images with snow in the background are predicted to be a wolf, and those with no snow in the background are predicted to be a husky, regardless of which species the image contains.

This experiment demonstrates why XAI is needed if ML models are going to be accepted as being reliable. Even if a model scores well in testing we are not certain it will be reliable 'in the field', as it may rely on dubious reasoning which the test data failed to expose.

The central idea behind LIME is to provide an interpretation for the prediction of a point via a local "explainer" model

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constructed around the point. Explainer models are easier to interpret and provide relative scores for the importance of input features in the determination of the output.

There have been numerous extensions to LIME [3][4][5] attempting to improve the explanations provided through better sampling points for the explainer model. One such extension is UnRAvEL (Uncertainty driven Robust Active Learning Based Locally Faithful Explanations) [6] which uses a novel Gaussian-Process based acquisition function called FUR (Faithful Uncertainty Reduction) to select points for the explainer model.

In early XAI work there was an emphasis for explainable models to provide feature importance scores to potential users and domain experts. This is intended to give the experts confidence that the black box model is making predictions for the right reasons, and to persuade them to trust the model outcomes.

Expanding the subjective concepts of confidence and trust to that of reliability will be a challenging task. One potential route is in the use of recognised metrics to give objective performance measurements of explainable models. A recent review [7] of XAI metrics has identified 12 metrics that consistently appear in papers on XAI. In the review they are given the name the "Co-12" properties (because the properties all have names beginning with "co").

In this paper we aim to demonstrate that we can use UnRAvEL to generate a set of explanations that represent a trade-off in specific Co-12 properties. In particular, we propose that by weighting the two terms within the FUR acquisition function used within UnRAvEL, different points will be sampled for the explainer model and thus different explanations provided. By controlling the weights, we can control (and ultimately optimize) the trade-off in Co-12 properties for an explanation.

The remainder of this paper is structured as follows. In Section 2 we introduce LIME and UnRAvEL. In Section 3 we summarize the Co-12 properties used to evaluate explanations. In Section 4 we present our methodology for generating different explanations using UnRAvEL. In Section 5 we present the results of our experiments and in Section 6 we summarize our findings.

2 Local Explainable Models

LIME

Figure 1 showed a black box model with its prediction f(X) being a function of the input features. LIME uses a local linear model to determine the importance of the input features. The prediction function f(X) for a linear model is given in Equation 1.

$$\boldsymbol{f(X) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n}$$

Equation 1

In a linear model each input feature is multiplied by a weight to calculate the prediction. This is a simple model to interpret, as the magnitude of a feature's weight determines its

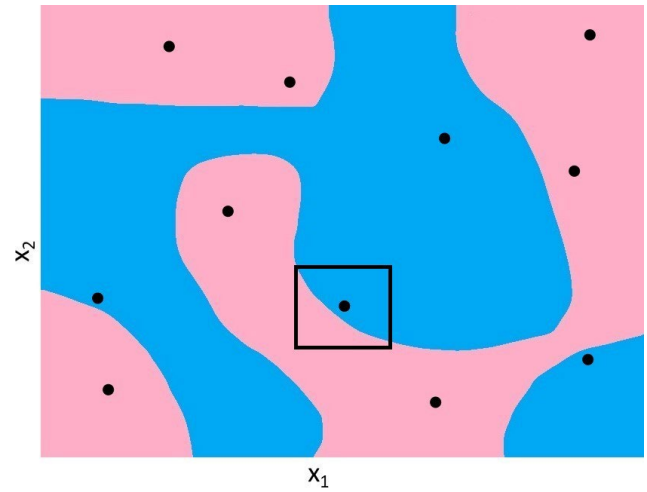

Figure 3. Two-Dimensional feature space with complex model prediction

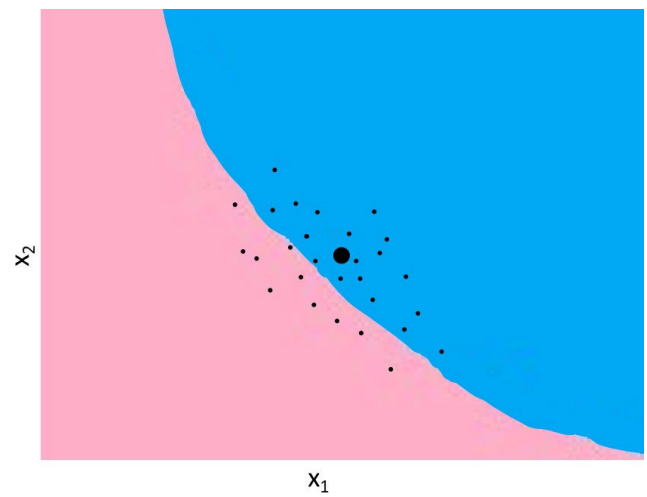

Figure 4. LIME creates new sample points around the point being explained

importance in the prediction. LIME uses linear models to locally interpret single points.

Figure 3 shows a 2-D feature space with features x_1 and x_2. A black box model in this feature space makes a prediction of either pink or blue. It appears to be complex model whose output is not easily interpretable.

The principle behind LIME is to focus on a single point, which is identified by the black square in Figure 3. In Figure 4 we zoom in on this point and create additional points around the point being explained. These points are generated from perturbations of the original point, and their predictions are given by the points being input to the black box model.

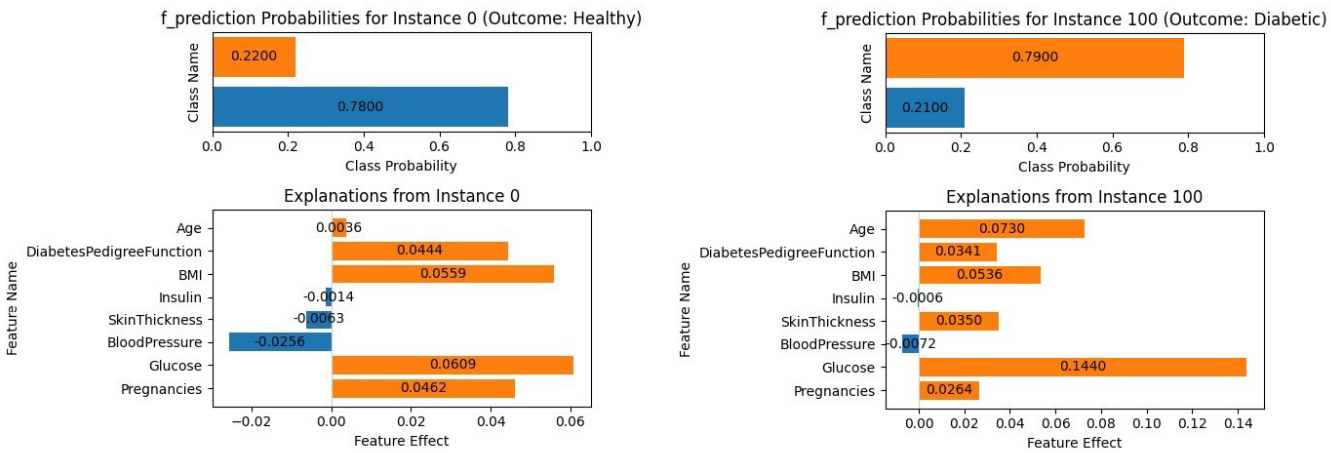

Figure 5. Predictions and feature importances for two instances of the diabetes dataset

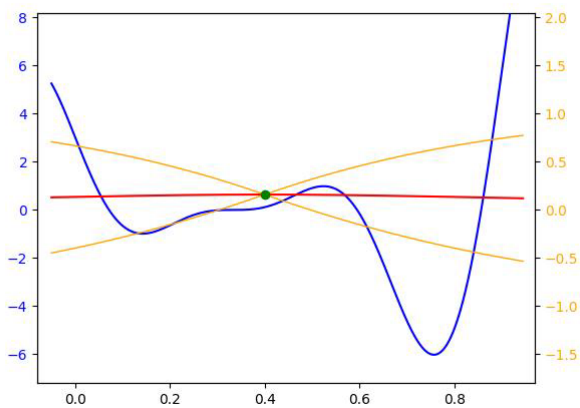

Figure 6. Gaussian Process modelling a 1-D Forrester function at first iteration

Using these additional points and their predictions a linear model is created to model behaviour around the point.

From Equation 1 we can see that the larger the weight for a feature the greater effect it has on the outcome. Therefore, weights in the local linear model give the importances of the features in the locality.

Figure 5 shows the predictions and feature importances for two points from the diabetes dataset use in this project. The feature importances were generated by a LIME model to explain those points. The importance corresponds to the weights of the linear model. Weights in a linear model can be positive (orange) or negative (blue).

UnRAvEL

UnRAvEL uses a Gaussian Process [8] to create a local interpretable model. Figure 6, Figure 7 and Figure 8 show the Gaussian Process (GP) modelling a 1-D Forrester function over a number if iterations. The Original function is the blue

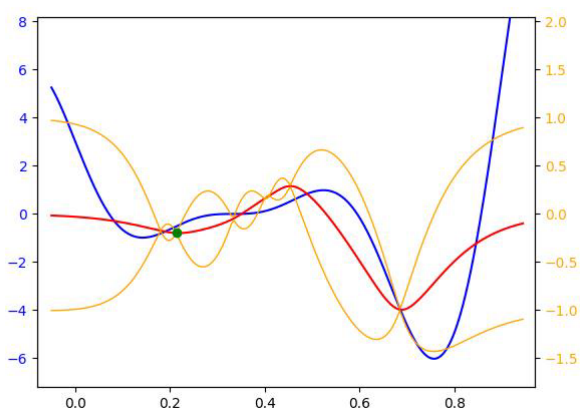

Figure 7. Gaussian Process modelling a 1-D Forrester function at 6th iteration

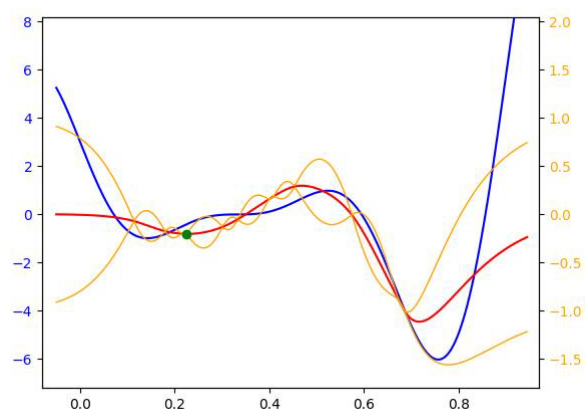

Figure 8. Gaussian Process modelling a 1-D Forrester function at final iteration

Table 1. Model Properties used to Access Performance

Property	Description
Correctness	Measures how faithfully the explainer model replicates the behaviour of the black box model. E.g. are the same features as important to both models?
Output-Completeness	This property looks at output of the explainer model compared to the black box model and measures the accuracy of the explainer model predictions.
Reasoning-Completeness	Measures how much the black box and explainer models are in step with their reasoning. E.g. do changes to input features have the same effect on the predictions of both models?
Consistency	The measures the consistency of explainer models when modelling the same feature space with each being based on different black box models.
Continuity	The measures the resilience of explainer models to perturbations of the features.
Jaccard Similarity	Measures the consistency of feature importance across all points in feature space.
Calibration	Calibration is a measure of well a model makes it predictions. It gauges how overconfident or under confident a model is performing.

line, the GP model is the red line, the orange lines represent the uncertainty in the GP prediction. In these figures the GP only models the function in the range 0.1 to 0.7, it does not select points outside this range.

Gaussian Processes use an iterative approach to create models, they select a new point to add to the model at each iteration. As points are added to the model its predictions become more accurate and the uncertainties diminish. We reach a point where adding more points does not significantly improve the model.

Figure 6 is the first iteration when only a single point, the dataset point being explained, has been added to the model. Figure 7 is when five addition points have been selected, and Figure 8 is the final iteration, when the model is complete.

A Gaussian Process is significantly more complex than linear models, but we can obtain feature importances from the model, which is essential in XAI.

Each iteration of a Gaussian Process model uses an acquisition function to select the next point for the model, UnRAvEL uses a novel acquisition function called FUR.

The choice of acquisition function significantly affects the performance of the GP model, and investigating UnRAvEL and Gaussian Process models was the route the project took rather than considering LIME. The main part of the project involved reconfiguring the FUR acquisition function to improve model performance.

3 Quantifying XAI Performance

The paper from Meike Nauta et al [7] is a detailed and far-reaching literature review of the current methods used to assess the performance of XAI models. The paper identifies twelve properties (referred to as the Co-12 properties) of XAI models that appear repeatedly in the reviewed papers. The paper looks to define common methods that are used across the reviewed set of papers that can be used determine these properties.

Five Co-12 properties have been chosen for this project, ones that can be objectively determined and can provide a measure of model performance. These are described in Table 1. There are two additional important metrics calculated that are not Co-12 properties. These are Jaccard score and calibration.

4 Methodology

FUR and FUR_W

In the UnRAvEL process when a data point is selected for explaining a Gaussian process model is created around that point. Addition points are incrementally added to the Gaussian process model, and the selection of these points is determined by the FUR acquisition function.

In the process the first point added to the GP is the point that is being explained. To determine the next and subsequent points the FUR [6] acquisition function is used. Acquisition functions use the values of the feature space, $X = (x_1, x_2, \ldots, x_n)$, as input. A search is made in the feature space around the point to determine the point that has the maximum value of FUR. This point is added to the model for the next iteration. The selection of the next point can be written:

$$X_{next} = \arg\max(T_1 + T_2)$$

Equation 2

Where:

$$T_1 = \|X - X_0 - \bar{\sigma}\epsilon/\log(n)\|_2$$

$$T_2 = \sigma_n(X)$$

- X_0 is the original point.
- $\bar{\sigma}$ is the mean of the standard deviation in the training data.
- n is the iteration number.

- ϵ is a noise factor to move the selected point away from X_0. Dividing this noise factor by log(n) reduces this noise as the iterations progress, this brings selected values closer to X_0.
- $\sigma_n(X)$ is the uncertainty in the model prediction at point X.

Figure 9 shows the values of terms T_1 and T_2 along with their combined value in FUR (right axis) in a feature space $0 \leq x \leq 1$, where the 'model' is the 1-D Forrester function (left axis).

The two terms T_1 and T_2 in the FUR acquisition function have different aims. T_1 is intended to make the selected points be located around the vicinity of the original point, initially from a wide range, but this range is reduced during later iterations, by dividing by log(n).

T_2 selects points where the uncertainty of the Gaussian process is greatest, with the aim of reducing this uncertainty.

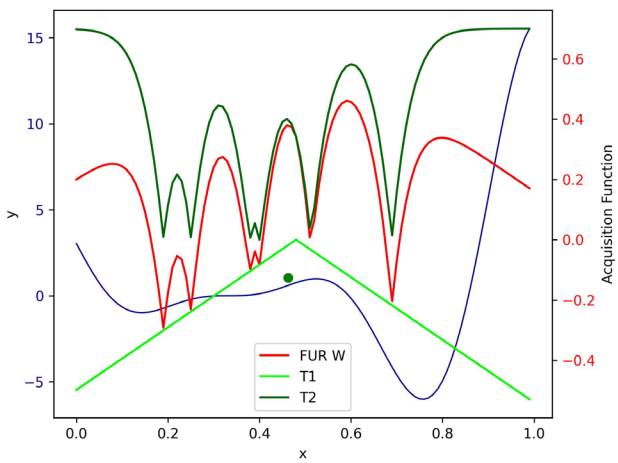

Figure 9. 1-D Forrester Function showing the terms T_1, T_2 from the FUR acquisition function

This project is to investigate these competing terms in the FUR acquisition function by introducing weights for each of the terms. We create a new acquisition function FUR_W, a weighted version of FUR. This is defined in Equation 3 below, where w is a weight between 0 and 1:

$$X_{next} = \arg\max(-wT_1 + (1-w)T_2)$$

Equation 3

Running Experiments

To investigate the effects of weighting the terms in FUR_W several datasets were used in each experiment.

Each dataset was used to train two black box models, a neural-network and random forest. Explainer models were then created to model and explain test points from the dataset.

A number UnRAvEL explainer models were generated for each black box model. Each explainer model was generated using a different weight value in its FUR_W functions. The weights used were 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95). The effect on the terms T_1 and T_2 in the FUR_W acquisition function can be seen in Figure 10 where T_2 is dominant, and Figure 11 where T_1 is dominant.

The performance of the explainer models was then measured using the selected properties in Table 1.

5 Results

One very noticeable property of the UnRAvEL explainer models is how accurate they are in their outcome predictions (the Output-Completeness property). Their predictions match closely the black box predictions, as shown in Figure 12. From this graph it appears the explainer model predictions are indistinguishable from those of the black box models.

A possible reason for this is that and there could be of over training in the explainer models. This is good when the explainer models are making predictions, but it could mean other properties of the explainer model will not score as well.

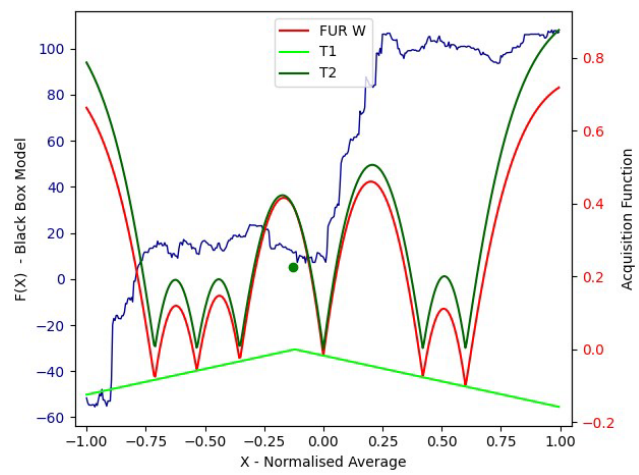

Figure 10. FUR_W acquisition function for weights of 0.1 for T1 and, 0.9 for T2. The dataset is synthetic regression data.

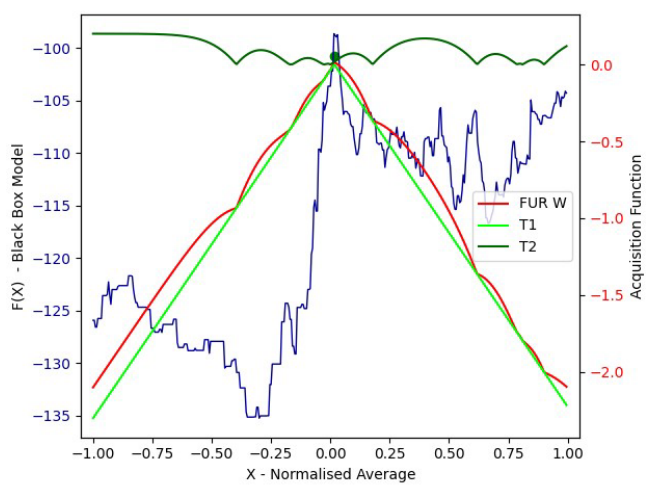

Figure 11. FUR_W acquisition function for weights of 0.8 for T1 and, 0.2 for T2. The dataset is synthetic regression data.

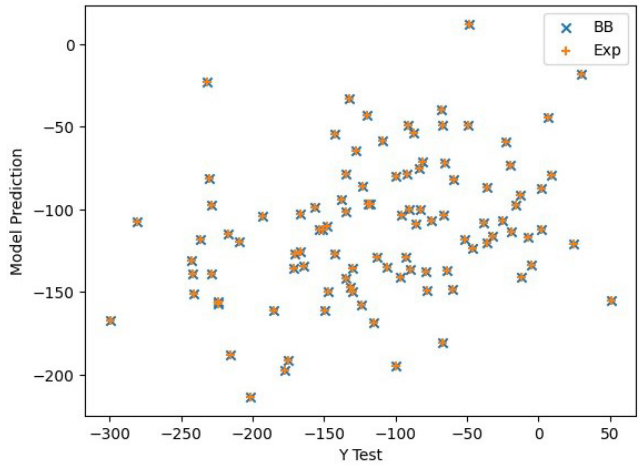

Figure 12. Comparison of predicted outcomes from black box model and explainer model for synthetic regression dataset.

Figure 13 shows the measurement of the Consistency metric for all datasets. This measures how well the predictions of two explainer models are correlated when they are trained on different types of black box models trained using the same dataset.

Other than the diabetes dataset there is high correlation between models, particularly for higher weight values. This indicates that the explainer models are modelling the dataset well at the individual points.

Unfortunately, many of the results resemble the graph in Figure 14. They show no clear patterns for the Co-12 properties as a result of varying the weights in the FUR_R function.

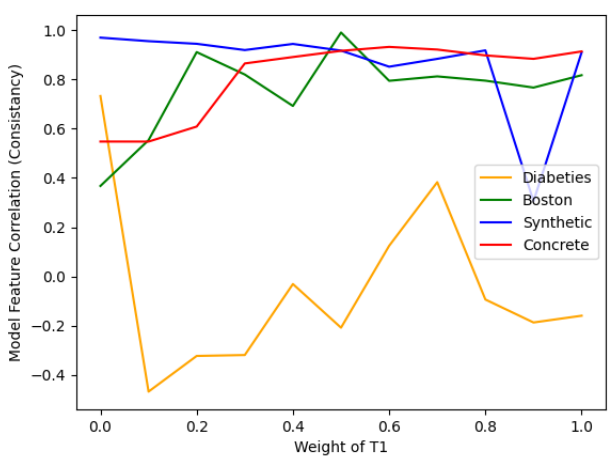

Figure 13. Consistency metric for all datasets.

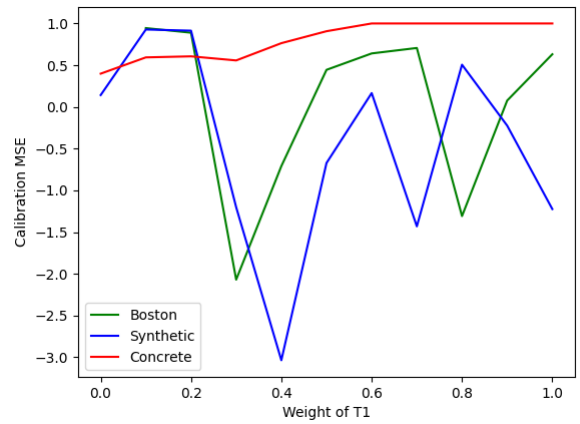

Figure 14. Correlation metric for all datasets

6 Conclusion

The models rate very highly in replicating the predictions of the black box models but less so on other metrics. As previously stated, the closeness in replicating black box model predictions may be a sign of overtraining the explainer model, the uncertainty values associated with the predictions are also very small (around 1e-5), which also indicates overtraining.

It would be interesting to reduce the number of points that are used to train the model to see if the other metrics could be improved upon without too great a loss of accuracy.

No results contain any clear patterns of behaviour that can be attributed to the settings of the weights. Drawing conclusions from the results on the effect of changing the weight of the terms in FUR_W is difficult.

Despite this apparent lack of success there have been benefits to the project. Looking at the Co-12 properties and how to implement their determination has been challenging. This work by itself could be the basis for future research, in developing better algorithms to calculate the Co-12 properties.

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